Chusong Zhang

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Education

Zhejiang University (U.S.News 35, 2026–2027) Zhejiang, China
Bachelor of Engineering in Automation, ***GPA: 4.04/4.3 (89.03/100)***
Minor in Advanced Honor Class of Engineering Education of Chu Kochen Honors College *Sep. 2022 – Jun. 2026*

The Hong Kong Polytechnic University Hong Kong, China
Exchange Student for Summer research
Advisor: Prof. LIN Wanyu, Department of Computing *Jul. 2025 – Oct. 2025*

Interests

Probabilistic Methods, Zero-shot Generation, Inverse Problems, PDE, UQ, Dynamics, GNN

Publications

- **Chusong Zhang**. (2026). "Restoring the Right Measure: Posterior-Consistent Flow Matching for PDE Inverse Problems", (In preparation, arXiv preprint)
- **Chusong Zhang**. (2026). "Asymptotic Inconsistency of Interpolation in Terminal-Projected Flows for PDE Inversion", (Under Review, arXiv preprint)
- H. Hong, W. Lin, **Chusong Zhang**, and K. C. Tan. "Geometric Graph Neural Diffusion for Stable Molecular Dynamics Simulations." *International Conference on Learning Representations (ICLR)*, 2026. [Paper]
- L. Du, X. Zhou, M. Chen, **Chusong Zhang**, Z. Su, P. Cheng, J. Chen, and Z. Zhang. "SoK: Dataset Copyright Auditing in Machine Learning Systems." 2025 IEEE Symposium on Security and Privacy (SP). [Paper]

Research Trajectory

Zero-Shot Generative Modeling for Industrial Surface Defect Detection

Student Project Leader, Sun Youxian Research Group, Zhejiang University *Oct. 2023 – Dec. 2024*

- Led a research project on cross-category zero-shot defect detection for industrial inspection, generating synthetic samples for unseen defect categories to address limited training data; identified the limited information structure and poor fine-grained detail of conventional cGANs as key challenges.
- Inspired by ancestral sampling, designed and implemented a **pyramid-structured conditional GAN (PycGAN)** with a U-Net backbone, decomposing defect generation across multiple resolutions from coarse geometric structure to fine-grained texture.
- Developed a hierarchical prior propagation mechanism in which the generated output at each resolution is resampled and propagated as an additional spatial condition to the next finer level; newly introduced high-resolution texture information is concatenated with the propagated prior and injected through cross-attention.
- Formulated the pyramid generation process as a hierarchical conditional decomposition of the generation distribution, mitigating mode collapse and reducing FID by 73.1% relative to a standard cGAN, with only a modest decrease in generation efficiency.

Control-Space Projection for Zero-Shot Constrained Diffusion

Independent Research *Feb. 2026 – May. 2026*

- Investigated zero-shot constrained generation, where gradient-based guidance provides limited constraint guarantees while state-space projection can disrupt the generative trajectory by repeatedly forcing intermediate states onto the constraint manifold.
- Proposed **PMP-Proj**, reformulating diffusion generation as a Hamiltonian control system and shifting constraint intervention from the generated state to the control variables via the Pontryagin Maximum Principle (PMP).
- Using a local optimal-control approximation, reduced each intervention step to a minimum-control-perturbation problem that steers the next state into the feasible region; further employed Lie derivatives to transform nonlinear, coupled constraints into a locally convex linear half-space in the control space.
- Achieved strong constraint satisfaction with $\mathcal{O}(1)$ intervention overhead, while preserving generation quality; demonstrated improvements over existing constraint-guidance and projection baselines on incompressible fluid fields and image inverse problems.

Asymptotic Inconsistency of Interpolation in Terminal-Projected Flows

Independent Research *Jun. 2026 – Sep. 2026*

- Investigated zero-shot hard-constrained PDE methods based on terminal projection, such as PCFM and ECI, focusing on whether their default reverse OT state interpolation assumption remains valid after terminal projection: whether the instantaneous generative velocity asymptotically recovers the reverse optimal-transport secant velocity as the step size $\Delta t \to 0$.
- Constructed the **Minimal Step-wise Perturbation (MSP)** framework as an analytical reference, which minimizes perturbation to the unconstrained next state subject to terminal feasibility, while avoiding reliance on the above asymptotic consistency assumption through only a correction of the projection center.
- Compared the models and proved that terminal projection breaks the above asymptotic consistency: reverse OT interpolation produces a $\Theta(1)$ state discrepancy and induces a surplus velocity field of order $\Theta(\Delta t^{-1})$; further proved that this velocity lacks posterior-guidance capability and causes trajectory deviation and posterior measure distortion.
- Validated the erroneous assumption and its resulting deficiencies in the original methods across multiple benchmarks. Found that performance is strongly affected by the previously unrecognized surplus velocity field, with incidental performance gains potentially arising from accidental alignment between the surplus direction and the posterior distribution, rather than from reliable posterior guidance.

Posterior-Consistent Flow Matching for Zero-Shot Constrained PDE Inversion

Independent Research *Sep. 2026 – Oct. 2026*

- Investigated posterior distortion in zero-shot hard-constrained generation, identifying that standard projection methods fail to capture the geometric co-area factor over zero-measure manifolds, causing severe posterior variance collapse and mode displacement.
- Proposed the **Conditional Minimal Step-Wise Perturbation** (CMSP) framework, reformulating constrained generation as conditional transport and inducing a Bayes-optimal conditional vector field directly from a pretrained flow to redirect the terminal attractor toward the true conditional posterior.
- Developed **Pre-Image Invariant Spectral Transfer** (PIS), leveraging an exact algebraic cancellation in the flow's Schur complement to achieve matrix-free velocity evaluation, reducing spatial memory complexity to $\mathcal{O}(d+m)$.
- Guaranteed exact hard constraint satisfaction and trajectory dynamical consistency without retraining; set new state-of-the-art accuracy in uncertainty quantification across geometric sweeps and high-dimensional PDE inverse problems.

Research and manuscript in preparation

CMSP-fast: From Iterative Trajectory Optimization to Source Correction

Oct. 2026 – Present

- By leveraging operator commutativity, we consolidate the complex iterative projection optimization steps inherent in **CMSP** into homogeneous operators. This insight leads to a highly efficient framework: a source-point correction followed by unconstrained flow propagation, concluded with a lightweight post-hoc projection. This approach accelerates the solving process while preserving **CMSP**'s advantage in Bayesian posterior consistency.

Extend PMP-Proj and Variational Control Perspective to Inverse Problems

Oct. 2026 – Present

- Reformulated diffusion-based inverse problems as a path-space variational control problem, deriving a dynamics-aware proximal guidance mechanism that integrates observation consistency with minimum-intervention principles. Developed a unified measurement-space optimization framework with provable local stability properties.

Honors and Awards

First Prize, National College Mathematics Competition (Top 1%)
Nandu First-class Scholarship (1 recipient)
Second Prize, National College Physics Competition
New Oriental Scholarship (10 recipients)
Departmental Top 10 Outstanding Student Award
First-class Scholarship, Zhejiang University